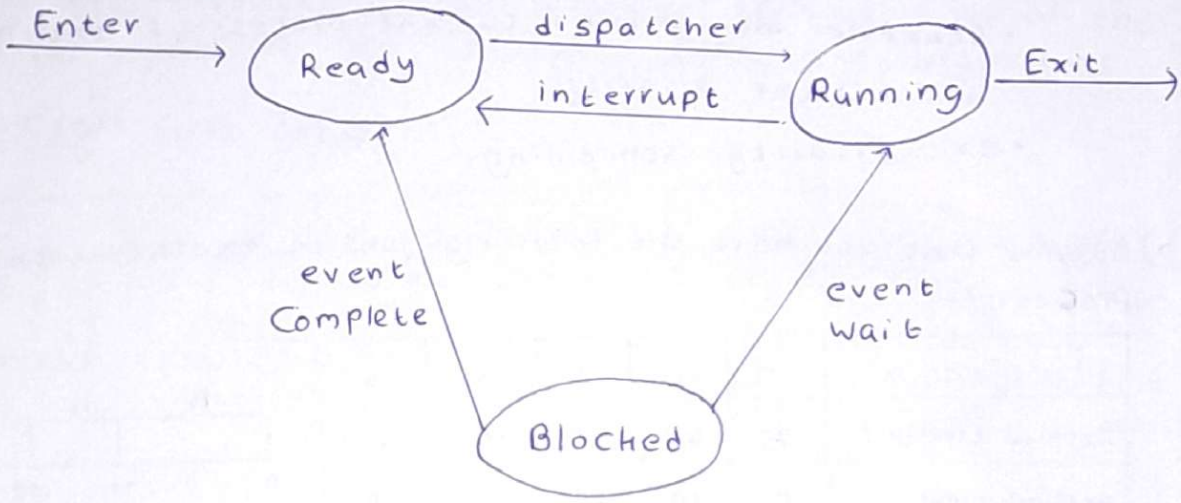
Tutorial 01
IT2244
2021/ICT/119

01) The multi programming system with a single processor (CPU) uses scheduling technologies to assign the processor (CPU) to processes.

a) State clearly the possible transition in a three-state process model with the aid of a diagram. In addition, indicate situations which would be used for scheduling algorithms.

Multiprogramming allows multiple processes to reside in memory and share the CPU. However, only one process executes at a time since there is only one CPU. The operating system uses scheduling algorithms to manage and assign the CPU. to different processes efficiently.



1. Ready → The process is prepared to run and is waiting for CPU.

2. Running → The process is currently being executed by the CPU

3. Blocked (waiting) → The process cannot continue until some I/O operation or event completes.

Situations that use scheduling Algorithms:

01. Multiple Ready Processes
- When multiple processes are ready to run, the Sch-eduler chooses which process to run next.
- ex:- After I/O event completes.

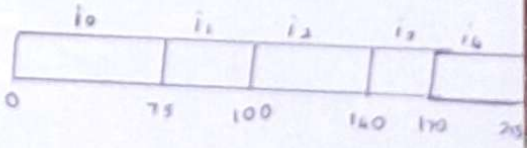
02. New process Arrival
- A newly created process is added to the ready queue.
- The scheduler decides whether the process run or wait. • ex:- FCFS, SJF

03. Time slice expiration
- If a running process uses up its time slice, it is moved back to the Ready process
- Scheduler selects another ready process to run next.
- ex:- Round Robbins

04. I/O completion
- A process moves from waiting to ready after I/O complete
- Scheduler may preept current process if the new one has higher priority.
- ex:- Priority scheduling.

b) Assume that you have the following jobs to execute with one Processor.

i	0	1	2	3	4
Service time(p_i)	75	40	25	30	45
Arrival time	0	10	10	55	95



i) Calculate the values for the following terms.

a. Service Rate $\rightarrow$ $\frac{\text{Number of jobs}}{\text{Total Service time}}$

$= \frac{5}{75+40+25+30+45}$

$= \frac{5}{215}$

≈ 0.023 jobs/unit time

B. Arrival Rate → $\frac{\text{No of jobs}}{\text{Total time span from first to last arrival}}$

$= \frac{5}{95-0} = \frac{5}{95}$

$= 0.0526$ jobs/unit time.

γ. Fraction of Busy time → $\frac{\text{Total service Time}}{\text{Finish time of Last job} - \text{Start Time of First Job.}}$

$= \frac{215}{215-0}$

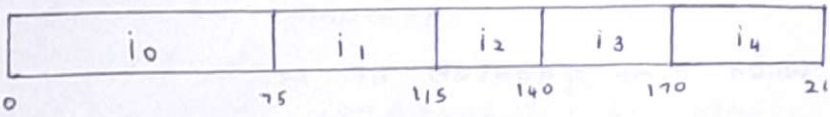
$= \frac{215}{215}$

$= 1.0$

• So CPU was busy 100% of time from the arrival of the first job to the finish of the last.

ii) Calculate the throughput and average waiting time of the system for each of the following scheduling algorithms.

∝. First Come First Serve



throughput = $\frac{\text{No of process}}{\text{Total Service Time}} = \frac{5}{215} = 0.023$ jobs/time unit

Average waiting time = $\frac{\text{Total waiting time}}{\text{No of Process}}$

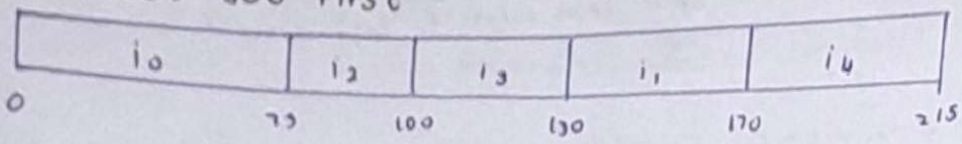
$= \frac{0 + (75-10) + (115-10) + (140-55) + (170+95)}{5}$

$= \frac{0+65+105+85+75}{5} = \frac{330}{5}$

$= 66$ time

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3. Shortest Job First



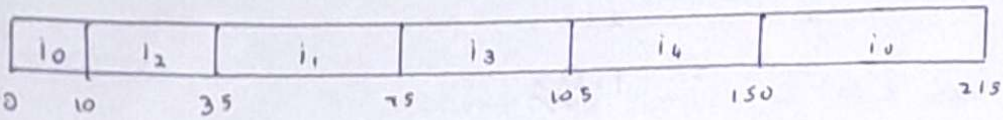
$$\text{throughput} = \frac{5}{215} = 0.023//$$

$$\text{Average waiting time} = \frac{0+65+45+120+75}{5}$$

$$= 61//$$

$i_0 \rightarrow 0$
$i_1 \rightarrow 130 - 10 = 120$
$i_2 \rightarrow 75 - 10 = 65$
$i_3 \rightarrow 100 - 55 = 45$
$i_4 \rightarrow 170 - 95 = 75$

4. Pre-emptive version of shortest job first.



$$\text{throughput} = 5/215 = 0.023//$$

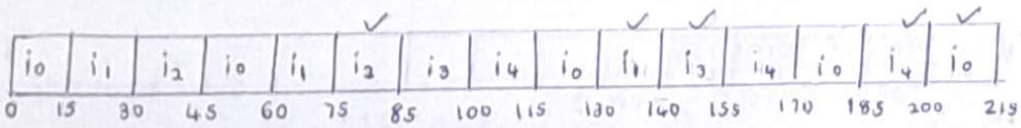
$$\text{Average waiting time} = \frac{140+25+0+20+10}{5}$$

$$= \frac{195}{5}$$

$$= 39//$$

$i_0 \rightarrow 140$
$i_1 \rightarrow 25$
$i_2 \rightarrow 0$
$i_3 \rightarrow 20$
$i_4 \rightarrow 10$

5. RR scheduling with time quantum of 15



$$\text{throughput} = 5/215 = 0.023//$$

$$\text{Average waiting time} = \frac{140+90+50+70+60}{5}$$

$$= \frac{410}{5}$$

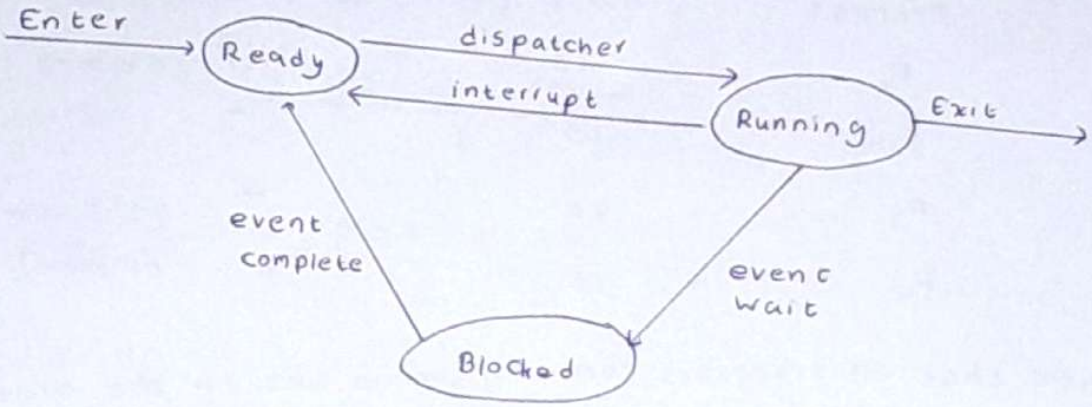
$$= 82//$$

$i_0 \rightarrow 215 - 75 = 140$
$i_1 \rightarrow 140 - 40 - 10 = 90$
$i_2 \rightarrow 85 - 25 - 10 = 50$
$i_3 \rightarrow 155 - 30 - 55 = 70$
$i_4 \rightarrow 200 - 45 - 95 = 60$

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02) A key concept in all Operating System is the process.

a) State clearly the possible transitions in a three state process model with aid of a diagram.



Transitions:

01) Enter → Ready

A new process is created and added to the ready queue.

02) Ready → Running

The dispatcher selects the process and assigns the CPU to it.

03) Running → Ready

The process is interrupted (eg:- time quantum expires in Round Robin) and returns to the ready queue.

04) Running → Blocked

The process requests an event (eg:- I/o operation) and waits for it to complete.

05) Blocked → Ready

The awaited event completes, and the process moves back to the ready queue.

06) Running → Exit

The process finishes execution and terminates.

Dispatcher → Decides which ready process gets the CPU (scheduling)

Interrupt → forces a running process to release the CPU (eg:- timer interrupt)

Event wait → Process pauses for I/O or other resources.

Event complete → Resource becomes available, unblocking the Process.

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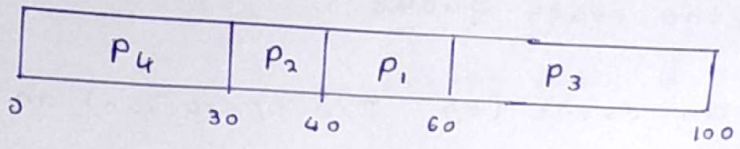
b) Consider the following set of processes with the CPU-burst times given in milliseconds as follows.

Process	Burst time	Priority
P_1	20	3
P_2	10	2
P_3	40	4
P_4	30	1 (Highest)

Assume that all processes come at time 0 but in the order P1, P2, P3 and P4 and also assume that the lower value in the Priority column indicates higher priority. You may take 5ms as a time quantum for Round-Robin scheduling.

Find out the Suitable Scheduling for three jobs, with the explanation.

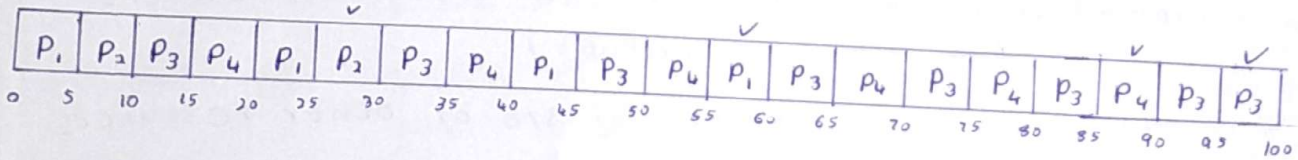
1. Priority order :- $P4 \rightarrow P2 \rightarrow P1 \rightarrow P3$



Avg completion time = $\frac{30+40+60+100}{4}$ = 57.5 ms //

Avg waiting time = $\frac{0+30+40+60}{4}$ = 32.5ms //

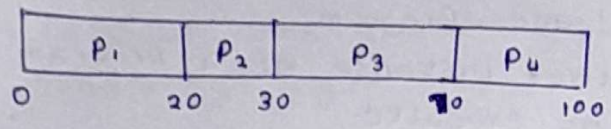
2. Round Robin



Avg comp time = $\frac{30+60+90+100}{4}$ = 70 ms //

Avg waiting time = $\frac{0+5+10+15}{4}$ = 7.5 ms //

3. FCFS



Avg completion time = $\frac{20+30+70+100}{4}$ = 55 ms //

Avg waiting time = $\frac{0+20+30+70}{4}$ = 30 ms //

1. Priority order

Pros – High priority jobs finished quickly

Cons - Starvation risk for low-priority jobs
If first job has low priority, the it has to wait long

Suitable for – Real-time systems where priority dictates urgency.

2. Round Robin

Pros - lowest avg waiting time comparing to other methods.
- no starvation

Cons - Higher completion time due to frequent context switches.

Suitable for - Time sharing systems.

3. FCFS

Pros - Simple
- No starvation

Cons - poor for short jobs
- If short job came late, It has to wait long time.

Suitable for - Batch systems where fairness is prioritized over responsiveness.

Comparment according to :-

- waiting time

① Round Robin (7.5ms)
② FCFS (30ms)
③ Priorty (32.5ms)

completion time

① FCFS (55ms)
② Priority (57.5ms)
③ Round Robin (70ms)

03)

a) Define the terms 'process' and 'program'

Process → A process is an active instance of a program that is currently being executed.

Program → A program is a set of instructions that tell a computer how to perform a task.

b) What is difference between "preemptive scheduling" and "Non preemptive scheduling".

Preemptive Scheduling	Non preemptive scheduling
• CPU can be taken from a process before it finishes.	• CPU cannot be taken until the process finishes.
• Process can be interrupted	• process runs till completion
• Better for urgent (high Priority) short tasks	• may delay short/high priority tasks.
• more frequent	• Less frequent/more stable.
• ex:- Round Robin	ex:- FCFS, SJF

c) Compare the advantages and disadvantages of the following Process scheduling algorithms.

i. Round Robin scheduling.

Advantages

- Every process gets equal time
- Good for multi-use and time sharing systems.
- No process waits forever (No starvation)

Disadvantage

- Too many switches if time slice is small.
- slow for long processes
- Average waiting time is not the best.

ii. Shortest Job First (SJF)

Advantages

- Fast overall performance
- Minimize average waiting time.
- Efficient for batch Processing

Disadvantage

- Long jobs may have to wait
- Needs knowledge of future burst times
- Can cause starvation of longer processes.

04)

a) Describe "Pre-emptive scheduling" and Non Preemptive scheduling" algorithms.

Pre-emptive Scheduling → The CPU can take control from a running process before it finishes and switch to another process.

ex:- Round Robin, Pre-emptive SJF

Non-Premptive Scheduling → Once the CPU is assigned to a process, it runs until completion of process It doesn't switch to another process until the current one is done.

ex:- FCFS, non Pre emptive SJF.

b) Discuss the advantages and disadvantages of the following scheduling algorithms,

i. Round Robin scheduling.

Advantages:

- Every process gets equal time
- Good for multi-user & time Sharing systems
- No process waits forever (No Starvation)

Disadvantages:-

- Too many switches if time slice is small.
- long burst time
- Slow for long process

ii. Priority scheduling:-

Advantages

- High priority tasks run first
- good for urgent job
- Efficent for real time systems.

Disadvantages.

- Unfair to low priority processes.
- May lead to Por avarage waiting time.

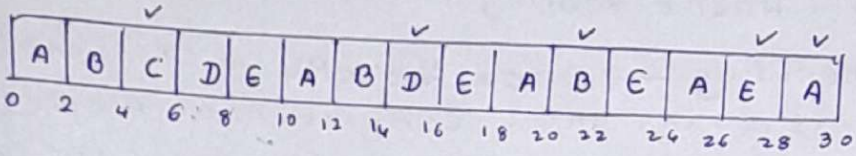
c) Five batch jobs A through E, arrive at a computer centre at almost the same time. They have estimated running times of 10, 6, 2, 4 and 8 minutes. Their (externally determinded) priorities are 3, 5, 2, 1 and 4 respectively, with 5 being the highest priority for each of the following scheduling algorithms, determine the mean process turnaround time. Ignore process-switching overhead.

<09>

1. Round robin

Job	A	B	C	D	E
running time	10	6	2	4	8
Priority	3	5 highest	2	1	4

i. Round robin

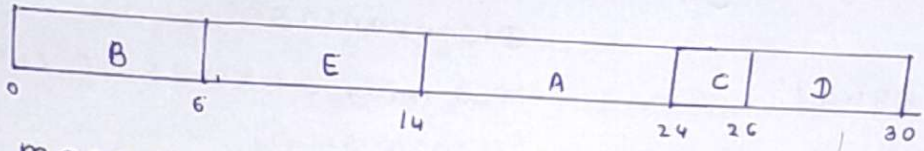


mean turn around time = $\frac{30+22+6+16+28}{5}$

= 20.4 min //

ii. Priority scheduling

B → E → A → C → D

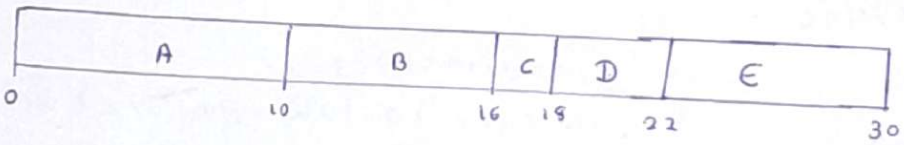


mean turn around time = $\frac{24+6+26+30+14}{5}$

= 20 min //

iii. FCFS

A → B → C → D → E

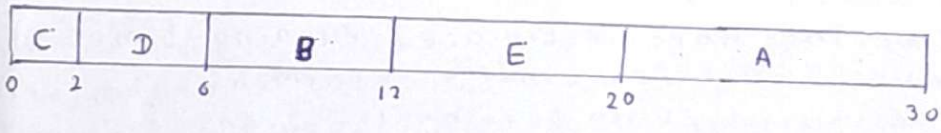


mean turn around time = $\frac{10+16+18+22+30}{5}$

= 19.2 min //

iv SJF

C → D → B → E → A



mean turn around time = $\frac{2+6+12+20+30}{5}$

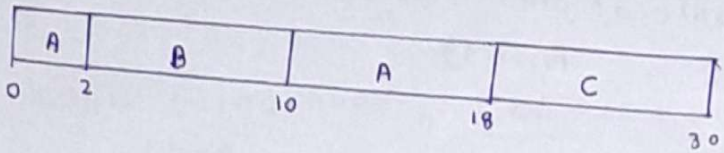
= 14 min //

* Best scheduling According to turn around time.

05)

a) For the following 3 jobs A, B, C what is the turnaround time for each job if pre-emptive shortest Job first algorithm is employed.

Job	Arrival time	CPU burst
A	0	10
B	2	8
C	4	12



Turnaround Time for each job

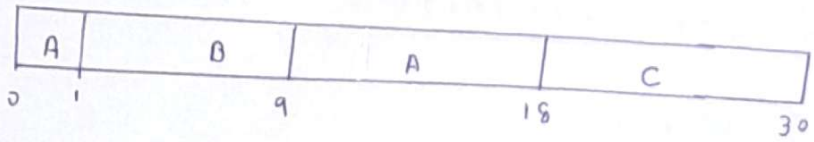
A → 18

B → 10 - 2 = 8

C → 30 - 4 = 26

b) For the following jobs A, B, C, draw the Gantt chart and calculate the average turnaround time employ Pre-emptive Priority scheduling.

Job	Arrival time	CPU burst	Priority
A	0	10	4
B	1	8	2
C	3	12	6

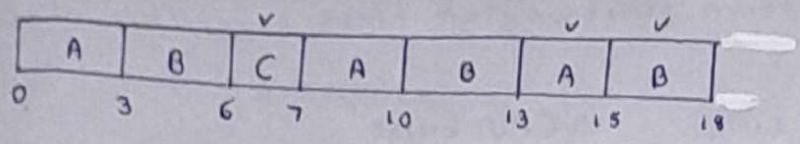


Average turnaround time $= \frac{18 + 8 + 27}{3}$

$= 17.67$ //

c) For the following list of jobs draw the Gantt chart and calculate the turn around time of each job when round robin scheduling is employed. The time slice is 3 units.

Job	A	B	C
CPU burst	8	9	1



Turnaround time for each

A ⟶ 15

B ⟶ 18

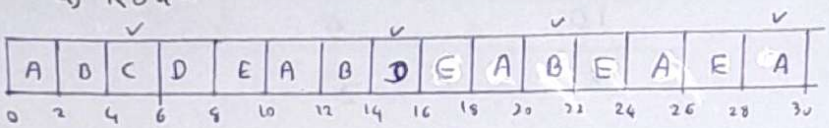
C ⟶ 7

06) Consider the following table of jobs

Job	CPU burst	Priority
A	10	3
B	6	5
C	2	2
D	4	1
E	8	4

For each of the following algorithms, determine the average turnaround time.

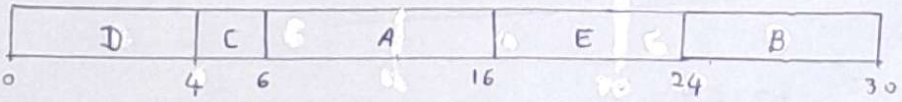
a) Round robin (The quantum - 2 units)



mean turn around time $= \frac{30+22+6+16+28}{5}$

$= 20.4$ units //

b) Priority scheduling

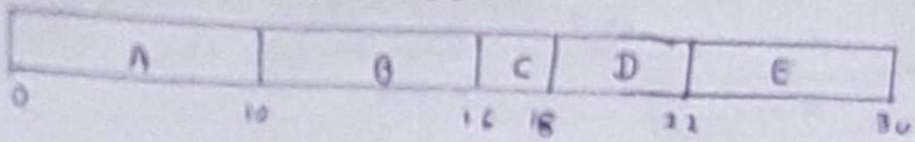


mean turn around time $= \frac{24+6+16+30+4}{5}$

$= 16$ units //

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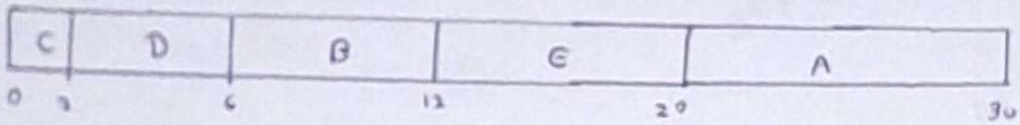
c) First - come first - serve



Mean turnaround time = $\frac{10+16+18+22+30}{5}$

= 19.2 units //

d) Shortest job first



Mean turnaround time = $\frac{2+6+12+20+30}{5}$

= 14 units //

07) Consider the following table of jobs

Job	A	B	C	D
CPU burst	6	3	8	2

Find the average turnaround time using the shortest job first scheduling algorithm.



Average turnaround time = $\frac{2+5+11+19}{4}$

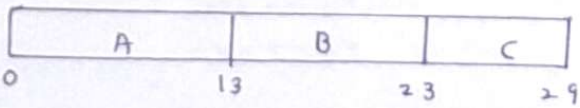
= 9.25 //

08) Consider the following table of jobs

Job	A	B	C
CPU burst	13	10	6

Draw the Gantt chart and find the average turnaround time for

a) FCFS scheduling algorithm



Turn around

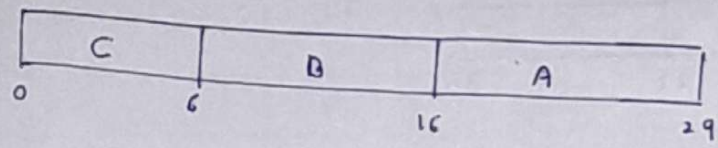
A → 13 - 0 = 13

B → 23 - 0 = 23

C → 29 - 0 = 29

Average turn-around time = $\frac{13+23+29}{3}$ = 21.67 //

b) SJF algorithm



A → 6
B → 16
C → 29

Average Turnaround Time $= \frac{6+16+29}{3}$

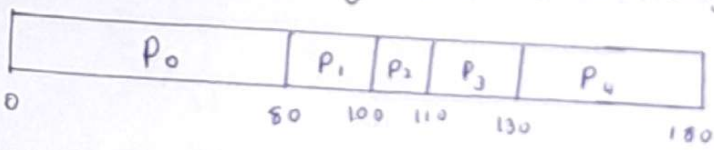
$= 17$ //

FCFS → 21.67
SJF → 17 r Best

09) Assume you have the following jobs to execute with one processor with the jobs arriving in the order listed here

P_i	0	1	2	3	4
T_i	80	20	10	20	50

a) Suppose a system uses FCFS scheduling, create a Gantt Chart illustrating the executing of these processes.



i. What is the turnaround time for process P_3

Turn around time $= 130 - 0 = 130$ //

ii. Estimate the wait time for process P_4

Waiting time = 130 //

iii. Calculate the average turnaround time for the processes

Average Turn around Time $= \frac{80+100+110+130+180}{5}$

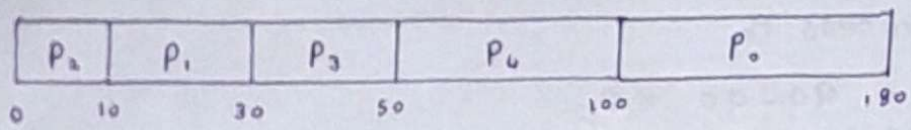
$= 600/5$

$= 120$ //

$P_0 \rightarrow 80$
$P_1 \rightarrow 100$
$P_2 \rightarrow 110$
$P_3 \rightarrow 130$
$P_4 \rightarrow 180$

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b) Suppose a system uses SJN scheduling, redo the questions (i) and and (ii) of part a.



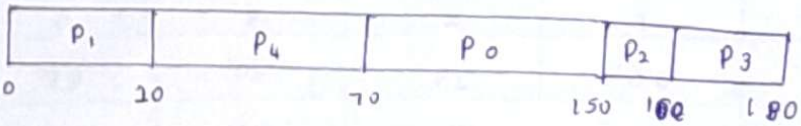
i) Turnaround Time for P3 = 50 - 0 = 50//

ii) wait time for P4 = 50//

(10) Consider the system that uses priority scheduling (under the following process load) where a small integer means a high priority.

Pi	0	1	2	3	4
Ti	80	20	10	20	50
Priority	3	1	4	5	2

a) Create a Gantt chart illustrating the execution of these processes.



b) What is turn around time process P_0

turnaround time = 150 - 0 = 150//

c) Calculate the average wait time for the processes

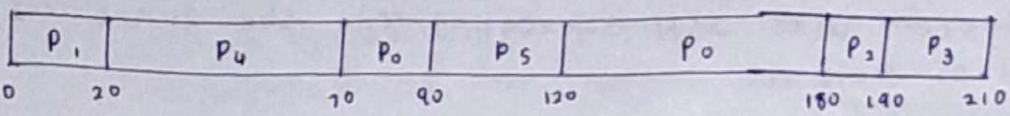
$P_0 \rightarrow 70$
$P_1 \rightarrow 0$
$P_2 \rightarrow 150$
$P_3 \rightarrow 160$
$P_4 \rightarrow 20$

Average waiting time $= \frac{70+0+150+160+20}{5}$

$= \frac{400}{5}$

$= 80//$

d) Suppose the process P5, service time of 30 arrives with the Priority of 2 at 90th unit of time and scheduling acts as pre-emptive, calculate the following values.



i. Wait time for process P_5

wait time = 90 − 90 = 0 //

ii. Turn around time for process P_0

Turnaround time = 180 − 0 = 180 //

iii. Average wait time for the processes.

$$\text{Average wait time} = \frac{100 + 0 + 180 + 190 + 20 + 0}{6}$$

$$= 81.67 //$$

$P_0 \rightarrow 70 + 30$
$P_1 \rightarrow 0$
$P_2 \rightarrow 180$
$P_3 \rightarrow 190$
$P_4 \rightarrow 20$
$P_5 \rightarrow 0$

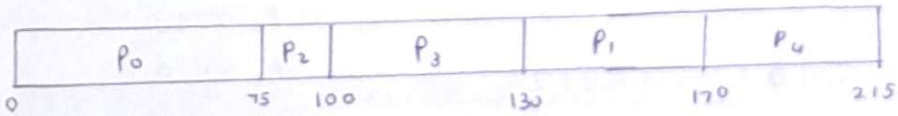
iv. Throughput

$$\text{throughput} = \frac{6}{210} = 0.0286 //$$

(ii) Assume you have the following jobs to execute with one Processor.

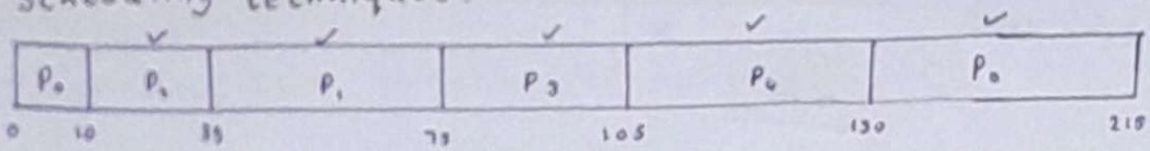
P_i	0	1	2	3	4
T_i	75	40	25	30	45
Arrival time	0	10	10	55	95

a) Suppose a system uses non-pre-emptive SJN scheduling. Calculate the throughput of the system



$$\text{Throughput} = \frac{5 \text{ jobs}}{215} = 0.0233 //$$

b) Suppose a system uses pre-emptive SJN scheduling, calculate the throughput of the system and further analyse the two scheduling techniques.



Throughput = 5/215 = 0.023//

Comparision:-

Non preemptive SJN average waiting time = (0 + 65 + 45 + 120 + 75) / 5

= 61//

Preemptive SJN average waiting time = (140 + 25 + 0 + 20 + 10) / 5

= 195/5 = 39//

Average Completion time = (215 + 75 + 130 + 100 + 170) / 5

= 138//

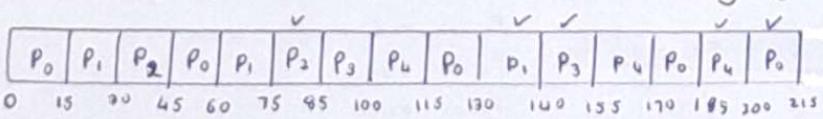
Average Completion time = (215 + 75 + 35 + 105 + 150) / 5

= 116//

method	Non Preemptive SJN	Preemptive SJN
Throughput	0.023	0.023
average waiting	61	39
average Completion	138	116

According to above 2 methods, Preemptive SJN is better than non Preemptive SJN because Preemptive SJN has lowest average waiting time and lowest completion time.

c) Suppose a system uses RR Scheduling with a quantum of 15 Create a Gantt chart illustrating execution of these processes.



i. Find the turnaround time for process P_3

turnaround time = Completion time - Arrival time

= 155 - 55

= 100//

ii) Calculate the average wait time for the processes.

$$\text{Average waiting time} = \frac{140+90+50+70+60}{5}$$

$$= \frac{410}{5}$$

$$= 82 //$$

iii) Calculate the average Turnaround time for the process.

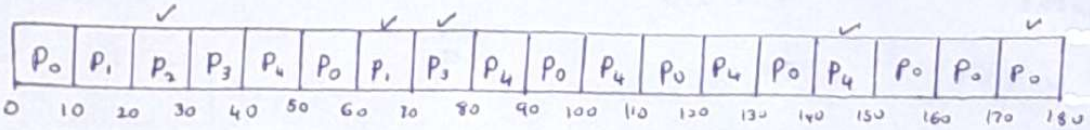
$P_0 \rightarrow 215 - 0 = 215$
$P_1 \rightarrow 140 - 10 = 130$
$P_2 \rightarrow 85 - 10 = 75$
$P_3 \rightarrow 155 - 55 = 100$
$P_4 \rightarrow 200 - 95 = 105$

$$\text{Average turnaround time} = \frac{215+130+75+100+105}{5}$$

$$= 125 //$$

(12) Consider the System that uses Round Robin Scheduling (Under the following process load)

P_i	0	1	2	3	4
T_i	80	20	10	20	50



a) Calculate the average wait time for the processes.

$P_0 \rightarrow 180 - 80 = 100$
$P_1 \rightarrow 70 - 20 = 50$
$P_2 \rightarrow 30 - 10 = 20$
$P_3 \rightarrow 80 - 20 = 60$
$P_4 \rightarrow 150 - 50 = 100$

$$\text{average waiting time} = \frac{100+50+20+60+100}{5}$$

$$= 66 //$$

b) Calculate the average Turn around time for the processes.

$P_0 \rightarrow 180$
$P_1 \rightarrow 70$
$P_2 \rightarrow 30$
$P_3 \rightarrow 80$
$P_4 \rightarrow 150$

$$\text{average turnaround time} = \frac{180+70+30+80+150}{5}$$

$$= \frac{510}{5}$$

$$= 102 //$$

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c) Describe the advantages and disadvantages of this scheduling technique.

Advantages :-

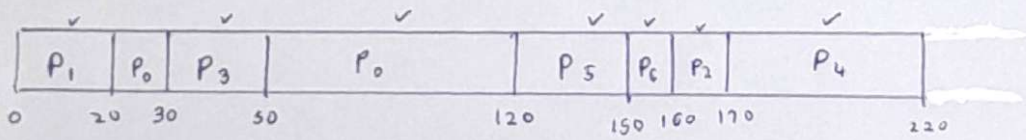
- Fairness : Each process gets equal CPU time slices.
- Prevents Stravation : All processes will eventually get Scheduled.
- Responsive : Good for time-sharing and interactive systems.

Disadvantages :-

- Small Processes get high completion time
- Long processes breaks lot of parts when time quantum is small.

(13) Consider the system that uses pre-emptive Priority Scheduling (under the following process load) where a small integer means a high priority.

P_i	0	1	2	3	4	5	6
T_i	80	20	10	20	50	30	10
Priority	2	1	3	1	4	2	1
Arrival time	0	0	10	30	95	100	150



i) Find the turnaround time for the process P_0

turnaround time = completion time - Arrival time

= 120 - 0

= 120 //

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ii) Calculate the average turnaround time and wait time for the processes.

$P_0 \rightarrow 120 - 0 = 120$
$P_1 \rightarrow 20 - 0 = 20$
$P_2 \rightarrow 170 - 10 = 160$
$P_3 \rightarrow 50 - 30 = 20$
$P_4 \rightarrow 220 - 95 = 125$
$P_5 \rightarrow 150 - 100 = 50$
$P_6 \rightarrow 160 - 150 = 10$

$$\text{average turnaround time} = \frac{120+20+160+20+125+50+10}{7}$$

$$= \frac{505}{7}$$

$$= 72.14 //$$

$P_0 \rightarrow 120 - 0 - 80 = 40$
$P_1 \rightarrow 20 - 0 - 20 = 0$
$P_2 \rightarrow 170 - 10 - 10 = 150$
$P_3 \rightarrow 50 - 30 - 20 = 0$
$P_4 \rightarrow 220 - 95 - 50 = 75$
$P_5 \rightarrow 150 - 100 - 30 = 20$
$P_6 \rightarrow 160 - 150 - 10 = 0$

$$\text{average wait time} = \frac{40+0+150+0+75+20+0}{7}$$

$$= 40.71 //$$

iii) with aid of above exercise, analyse this scheduling technique.

Preemptive Priority scheduling is good for high priority processes because highest priority processes get CPU access quickly (P_1, P_3, P_6) but even if shortest jobs get low priority it have to wait long time (P_2)

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